



TKR COLLEGE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS

**AI-POWERED REAL-TIME INDIAN
SIGN LANGUAGE RECOGNITION
AND BIDIRECTIONAL
COMMUNICATION SYSTEM WITH
LIVE SUBTITLE GENERATION**

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ABSTRACT

- Communication between hearing-impaired individuals and the general public is a major challenge in daily life.
- Indian Sign Language (ISL) is widely used by the deaf community, but many people do not understand it.
- Trained ISL interpreters are not always available, making communication difficult in schools, hospitals, workplaces, and public places.
- A real-time, intelligent, and accessible communication system is needed to bridge this gap.



INTRODUCTION

- Communication barriers between hearing-impaired individuals and the general public remain a significant challenge in daily life.
- Manual interpretation of Indian Sign Language (ISL) requires trained interpreters, who may not always be available.
- Existing sign language recognition systems often lack real-time performance, bidirectional communication, and support for Indian Sign Language.
- Artificial Intelligence (AI), Deep Learning (DL), and Computer Vision techniques enable accurate real-time recognition of ISL gestures and speech-to-text conversion.
- Our project aims to develop an **AI-Powered Real-Time Indian Sign Language Recognition and Bidirectional Communication System with Live Subtitle Generation**.
- The system translates Indian Sign Language into text and speech while converting spoken language into live subtitles, enabling seamless and inclusive communication between hearing-impaired and hearing individuals.



Existing system

- Existing systems use AI and Deep Learning for Indian Sign Language (ISL) recognition.
- They recognize hand gestures and convert them into text or speech.
- Most systems support only one-way communication.
- Commonly used models include CNN, LSTM, MediaPipe, and YOLO.
- They provide good accuracy but lack real-time bidirectional communication and live subtitle generation.



Limitation of Existing system

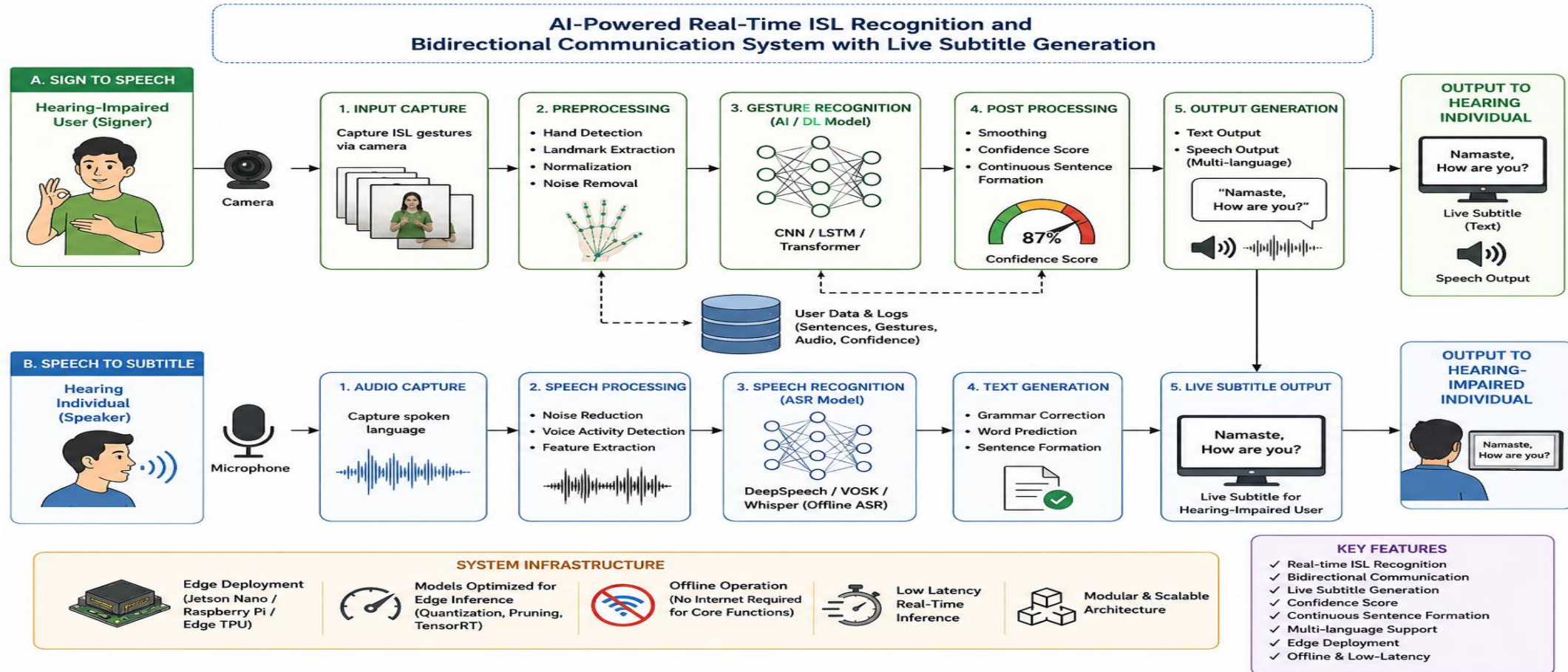
- Supports only one-way communication.
- Limited recognition of continuous sign language sentences.
- Accuracy decreases in complex backgrounds and poor lighting.
- Lacks real-time live subtitle generation.
- Limited support for Indian Sign Language (ISL) variations.
- High computational requirements in many existing models.



Proposed System

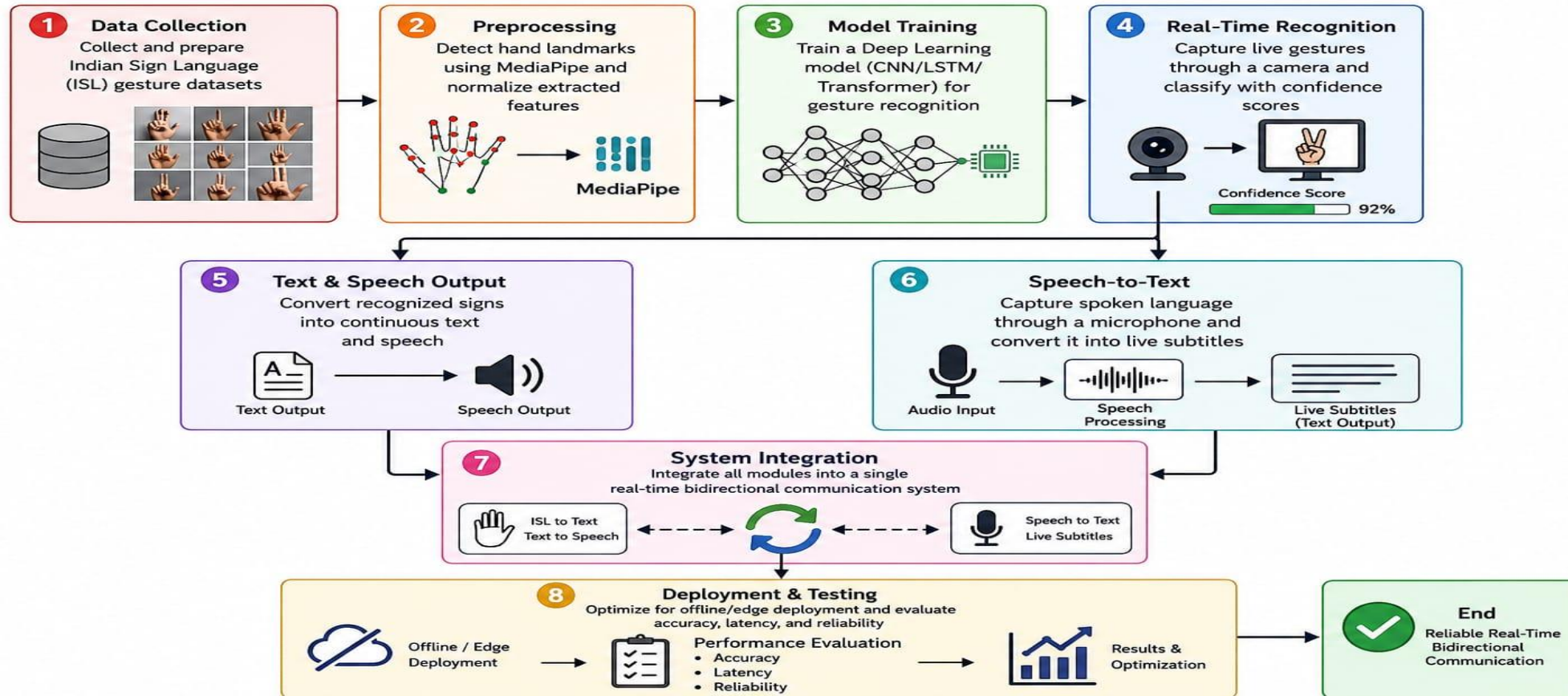
- Develops an AI-powered real-time Indian Sign Language (ISL) recognition system.
- Converts ISL gestures into text and speech instantly.
- Translates spoken language into live subtitles for bidirectional communication.
- Uses Deep Learning and Computer Vision for accurate gesture recognition.
- Provides fast, accurate, and user-friendly communication in real time.

ARCHITECTURE



METHODOLOGY

Indian Sign Language (ISL) Bidirectional Communication System Flowchart





METHODOLOGY

- **Data Collection:** Collect and prepare Indian Sign Language (ISL) gesture datasets.
- **Preprocessing:** Detect hand landmarks using **MediaPipe** and normalize the extracted features.
- **Model Training:** Train a **Deep Learning model (CNN/LSTM/Transformer)** for gesture recognition.
- **Real-Time Recognition:** Capture live gestures through a camera and classify them with confidence scores.
- **Text & Speech Output:** Convert recognized signs into **continuous text and speech**.
- **Speech-to-Text:** Capture spoken language through a microphone and convert it into **live subtitles**.
- **System Integration:** Integrate all modules into a single real-time bidirectional communication system.
- **Deployment & Testing:** Optimize for **offline/edge deployment** and evaluate accuracy, latency, and reliability.



SYSTEM REQUIREMENTS

Hardware Requirements:

Processor: Intel Core i5 or equivalent

RAM: 8 GB or higher

Camera: HD Webcam

Microphone: Built-in/External

Storage: 10 GB+ free space

Optional: Raspberry Pi / Jetson Nano for edge deployment

Software Requirements:

Operating System: Windows / Linux

Programming Language: Python

Libraries: OpenCV, MediaPipe, NumPy, TensorFlow/PyTorch

Speech Processing: VOSK / Whisper, pyttsx3

Interface: Streamlit / Custom GUI

Development Environment: VS Code / Jupyter Notebook

Database: SQLite (optional)



REQUIREMENTS

Functional Requirements

- Real-time ISL gesture recognition
- Gesture-to-text and text-to-speech conversion
- Speech-to-text with live subtitles
- Continuous sentence formation
- Confidence score generation
- Offline and low-latency operation
- Edge-device deployment support



CONCLUSION

- The proposed system enables **real-time bidirectional communication** by converting ISL gestures into text/speech and spoken language into live subtitles.
- It provides an **accessible, reliable, and user-friendly solution** to bridge communication gaps and promote inclusion in education, healthcare, workplaces, and public environments.
- The system reduces communication barriers by providing **fast and seamless interaction** between hearing-impaired and hearing individuals.
- Its **AI-based recognition, confidence scoring, and edge deployment** improve accuracy, reliability, and real-time performance.

*Thank
you*

